There are two important limits in this section,

$$\boxed{\lim_{x \to 0} \frac{\sin(x)}{x} = 1, \qquad \lim_{x \to 0} \frac{\cos(x) - 1}{x} = 0}$$

We will prove these later.

For now, we can use them to find $\frac{d}{dx}(\sin x)$

$$\lim_{h \to 0} \frac{\sin(x+h) - \sin(x)}{h} = \lim_{h \to 0} \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x)\cos(h) - \sin(x) + \cos(x)\sin(h)}{h}$$

$$= \lim_{h \to 0} \sin(x) \frac{\cos(h) - 1}{h} + \lim_{h \to 0} \cos(x) \frac{\sin(h)}{h}$$

$$= \sin(x) \cdot 0 + \cos(x) \cdot 1$$

$$= \cos(x)$$

So, $\boxed{\frac{d}{dx}(\sin(x)) = \cos(x)}$

One can similarly show $\boxed{\frac{d}{dx}(\cos(x)) = -\sin(x)}$

with these two derivatives, we can find all other trig derivatives using Quotient Rule

$$\frac{d}{dx}(\tan(x)) = \frac{d}{dx}\left(\frac{\sin(x)}{\cos(x)}\right) = \frac{\frac{d}{dx}(\sin(x))\cos(x) - \sin(x)\frac{d}{dx}(\cos(x))}{[\cos(x)]^2}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)}$$

Pythagorean Identity

$$= \frac{1}{\cos^2(x)} = \sec^2(x)$$

The Six trig derivatives are:

$$\boxed{\begin{aligned} &\frac{d}{dx}(\sin(x)) = \cos(x) && \frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x) \\ &\frac{d}{dx}(\cos(x)) = -\sin(x) && \frac{d}{dx}(\sec(x)) = \sec(x)\tan(x) \\ &\frac{d}{dx}(\tan(x)) = \sec^2(x) && \frac{d}{dx}(\cot(x)) = -\csc^2(x) \end{aligned}}$$

Ex: $F(x) = \dfrac{\sec(x)}{1+\tan(x)}$

$$f'(x) = \frac{\frac{d}{dx}(\sec(x))(1+\tan(x)) - \sec(x)\frac{d}{dx}(1+\tan(x))}{[1+\tan(x)]^2}$$

$$= \frac{\sec(x)\tan(x)(1+\tan(x)) - \sec(x)\sec^2(x)}{[1+\tan(x)]^2}$$

$$= \frac{\sec(x)\tan(x) + \sec(x)\tan^2(x) - \sec^3(x)}{[1+\tan(x)]^2}$$

$$= \frac{\sec(x)\left(\tan(x) + \tan^2(x) - \sec^2(x)\right)}{[1+\tan(x)]^2}$$

$\tan^2(x) = \sec^2(x) - 1$

$$= \frac{\sec(x)\left(\tan(x) - 1\right)}{[1+\tan(x)]^2}$$

Ex: Find the 27^{th} derivative of $\cos(x)$

$f(x) = \cos(x)$

$f'(x) = -\sin(x)$

$f''(x) = -\cos(x)$

$f'''(x) = \sin(x)$

$f^{(4)}(x) = \cos(x)$

$\vdots$

$f^{(4k)}(x) = \cos(x)$

$\downarrow$

$f^{(24)}(x) = \cos(x)$

$f^{(25)}(x) = -\sin(x)$

$f^{(26)}(x) = -\cos(x)$

$\underline{f^{(27)}(x) = \sin(x)}$

Special Trig limits

As mentioned earlier, the limits

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1, \qquad \lim_{x\to 0} \frac{\cos(x)-1}{x} = 0$$

are important. let's prove them

$\lim_{x\to 0} \frac{\sin(x)}{x} = 1$, we will show this via the squeeze thm.

<u>Claim:</u>

$$\cos(x) \leq \frac{\sin(x)}{x} \leq 1 \quad \text{for } x \in (0, \tfrac{\pi}{2})$$

Quadrant 1

rewriting the inequality,

$$\cos(x) \leq \frac{\sin(x)}{x} \leq 1$$

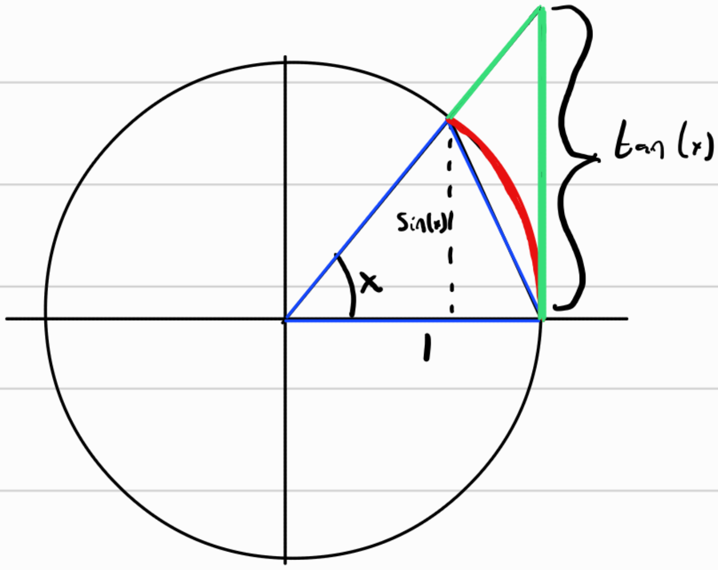
$$\frac{1}{\cos(x)} \geq \frac{x}{\sin(x)} \geq 1$$

$$\tan(x) \geq x \geq \sin(x)$$

$$\sin(x) \leq x \leq \tan(x) \quad \text{for } x \in (0, \tfrac{\pi}{2})$$

So, if this inequality is true, so is the one above.

On a unit Circle



The blue triangle has area A
The red Sector has area B
The green triangle has area C

We notice that $A \leq B \leq C$.

We Calculate

$$A = \frac{1}{2} \cdot 1 \cdot \sin(x) = \frac{1}{2}\sin(x)$$

$$B = (1) \cdot \frac{x}{2} = \frac{1}{2}x$$

$$C = \frac{1}{2}(1) \cdot \tan(x) = \frac{1}{2}\tan(x)$$

Sector area $= r^2 \cdot \frac{\theta}{2}$

So,

$$\frac{1}{2}\sin(x) \leq \frac{1}{2}x \leq \frac{1}{2}\tan(x)$$

$$\sin(x) \leq x \leq \tan(x) \qquad \text{for all } x \in (0, \pi/2)$$

So, the original inequality must be true. That is

$$\cos(x) \leq \frac{\sin(x)}{x} \leq 1 \quad \text{for all } x \in (0, \pi/2)$$

Then, by the squeeze theorem

$$\lim_{x \to 0} \cos(x) \leq \lim_{x \to 0} \frac{\sin(x)}{x} \leq \lim_{x \to 0} 1$$

$$1 \leq \lim_{x \to 0} \frac{\sin(x)}{x} \leq 1$$

So

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \quad \text{as wanted.}$$

Now, the other limit. Luckily this is simpiler,

$$\lim_{x \to 0} \frac{\cos(x) - 1}{x} \quad \frac{\cos(x)+1}{\cos(x)+1}$$

$$= \lim_{x \to 0} \frac{\cos^2(x) - 1}{x(\cos(x)+1)}$$

$$= \lim_{x \to 0} \frac{-\sin^2(x)}{x(\cos(x)+1)}$$

$$= \lim_{x \to 0} \frac{\sin(x)}{x} \cdot \lim_{x \to 0} \frac{-\sin(x)}{\cos(x)+1}$$

$$= 1 \cdot \frac{0}{2}$$

$$= 0 \quad \text{as wanted.}$$

Ex: $\lim_{x\to 0} \frac{\sin(7x)}{4x}$

$= \lim_{x\to 0} \frac{\sin(7x)}{4x} \cdot \frac{7/4}{7/4}$

$= \lim_{x\to 0} \frac{7}{4} \frac{\sin(7x)}{7x}$, let $\theta = 7x$, then as $x \to 0$, $\theta \to 0$ as well

$= \frac{7}{4} \lim_{\theta\to 0} \frac{\sin(\theta)}{\theta}$

$= \boxed{\frac{7}{4}}$

Ex: $\lim_{x\to 0} x\cot(x)$

$= \lim_{x\to 0} \frac{x\cos(x)}{\sin(x)}$

$= \lim_{x\to 0} \frac{\cos(x)}{\frac{\sin(x)}{x}}$

Cosine is continuous.

$= \frac{\lim_{x\to 0} \cos(x)}{\lim_{x\to 0} \frac{\sin(x)}{x}} = \frac{1}{1} = \boxed{1}$

Ex: $\lim_{\theta\to 0} \frac{\cos\theta - 1}{\sin\theta} \left(\frac{1/\theta}{1/\theta}\right)$

$= \lim_{\theta\to 0} \frac{\frac{\cos(\theta)-1}{\theta}}{\frac{\sin\theta}{\theta}} = \frac{0}{1} = \boxed{0}$